

Jessalyn Sebastian

Ph.D. Candidate at University of California, Irvine

Advisor: Volodymyr Minin

Research Interests: stochastic processes, Bayesian modeling, model robustness and prior sensitivity analysis, applications in infectious disease epidemiology, statistics education

Education

2024; M.S. Statistics, University of California, Irvine (UCI)

2021; B.S. Statistics and Data Science (*summa cum laude*), minor in Physics, University of California, Santa Barbara (UCSB)

Internships

2024; Centers for Disease Control and Prevention (CDC) Summer Internship in Forecasting and Outbreak Analytics, Nowcasting and Natural History team.

Publications

Peer-Reviewed

Sebastian, J.N., Minin, V.M. (2026). Gaussian Process Priors with Markov Properties for Effective Reproduction Number Estimation. *Journal of the Royal Statistical Society, Series A*. doi: 10.1093/jrssa/qnag088

Němcová, B., Goldstein, I.H., **Sebastian, J.N.**, Minin, V.M., Bracher, J. (2026). Unjustified Poisson assumptions lead to overconfident estimates of the effective reproductive number. *Epidemiology & Infection*, 154, e79. doi: 10.1017/S0950268826101605

Li, H., Zhou, M., **Sebastian, J.**, Wu, J., & Gu, M. (2022). Efficient Force Field and energy emulation through partition of permutationally equivalent atoms. *The Journal of Chemical Physics*, 156(18), 184304. doi: 10.1063/5.0088017

Under Review

Yek, C., **Sebastian, J.N.**, et al. (2025). Mapping respiratory virus circulation in Cambodia using metagenomic sequencing: a multi-center cross-sectional study.

Baniasad, A., Chao, S., Nguyen, J.A., Tian, E., Modongo, C., Minin, V.M., **Sebastian, J.N.**, and Shin, S.S. (2025). HIV Remains a Risk Factor for Unfavorable Tuberculosis Treatment Outcomes in the Era of Universal Access to Antiretroviral Therapy in Botswana. *medRxiv* preprint. doi: 10.1101/2025.11.26.25340699

Presentations and Posters

Locally Adaptive Smoothing with Subordinated Integrated Brownian Motion (Presentation). Joint Statistical Meetings, Nashville, TN, August 2025.

Gaussian Process Priors with Markov Properties for Effective Reproduction Number Inference (Presentation) The Western North American Region (WNAR) of The International Biometric Society, Whistler, BC, June 2025.

(Poster) 14th International Conference on Bayesian Nonparametrics, Los Angeles, CA, June 2025.

(Presentation) Joint Statistical Meetings, Portland, OR, August 2024.

Takeaways from Readings Selected by the Dogucu Research Group (Flash Talk). Electronic Conference on Teaching Statistics: Regional Conference at UC Irvine, Irvine, CA, June 2024.

The Ornstein-Uhlenbeck Prior for Effective Reproduction Number Estimation (Poster). UCI Data Science Initiative: Statistical and Machine Learning Applications in Biomedical Sciences Workshop, Irvine, CA, February 2024.

Software

epinowcast Tools to enable flexible and efficient hierarchical nowcasting of epidemiological time-series using a semi-mechanistic Bayesian model with support for a range of reporting and generative processes. (*author: contributed issues, code review, PRs*)
<https://github.com/epinowcast/epinowcast>

Teaching and Academic Mentoring

2026; Instructor of Record for:

STATS 8: Introduction to Biological Statistics

STATS 67: Introduction to Probability and Statistics for Computer Science, UCI Statistics Department

2025; Teaching Assistant for Introduction to Bayesian Statistics with Applications, International Centre for Mathematics in Ukraine

2025; Research Mentor for NIH Summer Institute for Biostatistics and Data Science (SIBS), UCI

2023; Instructor of Record for STATS 67: Introduction to Probability and Statistics for Computer Science, UCI Statistics Department

2021-2022, 2025-2026; Teaching Assistant for

STATS 211: Statistical Methods II: Generalized Linear Models

STATS 230: Statistical Computing

STATS 270: Stochastic Processes

STATS 7: Basic Statistics

STATS 8: Introduction to Biological Statistics, UCI Statistics Department

2020-2021; Learning Assistant for PSTAT 5A: Introduction to Statistics, UCSB Statistics and Applied Probability Department

2020-2021; Probability and Statistics Tutor for PSTAT 109: Statistics for Economics, UCSB Campus Learning Assistive Services

Service

2025-2026; Diverse Educational Community and Doctoral Experience (DECADE) Peer Mentor

2024-2026; Graduate Student Representative, UCI Statistics Department

2024-2025; Pacific Alliance for Low-income Inclusion in Statistics and Data Science (PALiISaDS) Mentor

2023-2026; Brown Bag Seminar Organizer, UCI Statistics Department

2023-2025; DECADE Student Representative, UCI Statistics Department

2017-2020; Community council member and Queer and Transgender Student Engagement Chair, UCSB Residential Housing Association

Reviewer for: Nature, Annals of Applied Statistics, PLOS Global Public Health

Honors and Achievements

2025; Runner-up for the Most Outstanding Written Paper Award, WNAR of The International Biometric Society

2025; Summer Scholar, UCI Department of Teaching Excellence and Innovation

2020; Phi Beta Kappa Society

2017-2021; UCSB Regents Scholarship

2017-2021; UCSB Dean's Honors

Programming Languages: R, Python, Stan, Julia, C++